FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Phishing –This refers to a form of fraudulent activity specifically aimed at attacking an email application to steal personal information. It occurs when a user is contacted by email or instant message OR IN ANY other way by masquerading as a legitimate person or organization. The intent is to trick the recipient into installing malware on their device or into sharing personal information, such as login credentials or financial information. An example is receiving a congratulatory email for winning a prize, the mail looks like it was sent from a known retail store and asks one to click on a link to claim the prize only to be redirected to a fake site that asks you to enter personal information or may even install a backdoor/virus on your device that monitors your activities.

Spear phishing-This is a highly targeted attack that sends customized emails to a specific person based on information the attacker knows about them-which could be their preferences, interests, activities and work projects. For example a cyber-attacker through their research discovers you are looking for a specific type of a house. They then join a real estate you are a member of, forges house offering and sends you an email that contains a link to see pictures of the house. After clicking the link unknowingly, you install a malware on your device or to other extend the give you an offer /discounted price only for you to be duped into buying a no-existent property.

KEY FEATURES

1. **IDENTIFYING PHISHING INDICATORS**
    - Suspicious sender address(spoofed sender)-Lookout for misspellings e.g go0gle.com instead of google.com
    - Urgency/threatening language-phrases such as immediate action, account warning suspension which are designed to make one action without thinking.
    - Generic greetings-Lack of customization of the email, like “dear sir/madam” can suggest mass phishing attempt.
    - Fake domains/suspicious links or attachment-before clicking on a link, hover the mouse to see the actual URL destination. They use tactics such as typosquatting e.g g00gle.com, subdomain tricks etc.
    - Requests for sensitive information such as passwords, bank details, credit card numbers. Legitimate companies rarely ask for these credentials.
    - Grammar and spelling errors are common red flags.

PREVENTION AGAINST PHISHING TECHNIQUES.

- **Inspect the URL:** Check that the real domain name immediately precedes the last dot (e.g., site.com, not site.com-secure.net).

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

- **Hover Before Clicking:** On desktop, hover over links to see the true destination URL in the corner of the browser.
- **Use Caution with Urgent Messages:** Phishing emails often create a sense of panic, demanding immediate login to "fix" an account.
- **Check for HTTPS/Security Certificates:** While many fake sites now use SSL, a missing secure lock or a suspicious certificate warning is a major red flag.
- **Verify Independently:** If an email seems suspicious, navigate to the official website directly through a search engine rather than clicking the link.
- **Stop and Think:** Be skeptical of unsolicited, high-pressure emails.
- **Verify Directly:** Contact the company directly through official channels, not through the link or phone number in the suspicious email.
- **Use Security Tools:** Keep your anti-virus software updated and use phishing filters provided by your email provider.

2. CLASSIFYING EMAILS AS SAFE, SUSPICIOUS AND PHISHING

Emails are classified as **Safe** (expected, authenticated sender), **Suspicious** (red flags, unusual content), or **Phishing** (malicious, trying to steal data) by analyzing sender addresses, urgency, links, and attachments. Safe emails have valid, known origins, while phishing attempts often include forged links, urgent demands, or malware. Classifying emails as **Safe, Suspicious, or Phishing** is a key part of email security and helps protect users from scams and malware. Here's a clear way to understand each category:

Safe Email:

- **Sender:** Verified, trusted, and familiar domain.
- **Content:** Expected, professional, with clear grammar.
- **Links/Attachments:** Secure, expected, and functional.

Example:

- A receipt from a service you actually used
- A normal message from your colleague or friend

Suspicious Email (Warning Flags):

- **Sender:** Mismatched, unexpected, or slightly altered domain names.

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

- **Tone:** High-pressure tactics, threats, or false urgency (e.g., "Account suspended", "Verify immediately").
- **Content:** Generic greetings like "Dear Customer" instead of your name.
- **Links/Attachments:** Unsolicited links, weird URLs, or unexpected .zip/.exe attachments.
- Unknown or slightly altered sender address (e.g., amaz0n.com instead of amazon.com)

Phishing Email (Active Threat):

- **Goal:** Attempts to steal sensitive information (passwords, banking details) or install malware.
- **Characteristics:** Impersonates well-known brands or colleagues, phishing for credentials via fake login pages.
- Pretend to be from trusted organizations (banks, tech companies)
- Create **urgency or fear** ("Your account will be suspended today!")
- Ask for sensitive data (passwords, PINs, OTPs)
- Contain fake links that mimic real websites
- May include malicious attachments

- **Links/Attachments:** Malicious URLs or files that, if clicked, download viruses or lead to fraudulent websites.

Example:

- "Click here to reset your bank password immediately or your account will be locked"

3. **EXPLAIN PHISHING TECHNIQUES IN SIMPLE, BUSINESS FRIENDLY LANGUAGE**
   Phishing is when attackers **pretend to be someone you trust** (like your bank, boss, or a well-known company) to trick you into **sharing sensitive information** or **clicking harmful links**.

VISHING

Often referred to as voice phishing, this type of attack sees criminals use voice communication technology to encourage users to divulge information, such as their credit card details.

Criminals can spoof phone calls using voice over internet protocol (VoIP), or leave recorded messages to give the impression that they are legitimate callers.

PHARMING

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

This type of attack deliberately misdirects users to a fake version of an official website. Tricked into believing that they are connected to a legitimate site, users enter their credentials into the fraudulent website.

WHALING

Whaling is a phishing attack that targets high profile individuals, such as senior executives within an organization, politicians and celebrities.

1. Impersonation (Pretending to be Someone Trusted)

   Attackers disguise themselves as:

- Your manager or CEO
- A bank or supplier
- A known company (e.g., Microsoft, PayPal)

Business example:
"Hi, this is the CEO—please urgently send me the latest payroll file."

Goal: Make you trust the message and act quickly.

2. Urgency and Pressure

They create panic so you don't think carefully.

Examples:

- "Your account will be locked in 1 hour"
- "Immediate payment required to avoid service disruption"

Goal: Force quick decisions without verification.

3. Fake Links (Look Real but Aren't)

Emails include links that appear legitimate but lead to fake websites.

Example:

- Looks like: www.bank.com

Goal: Steal login details when you enter them.

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

4. Malicious Attachments

Emails include infected files disguised as:

- Invoices
- Reports
- CVs or job applications

Goal: Install malware or ransomware when opened.

5. Credential Harvesting

You're asked to "log in" or "confirm details."

Examples:

- "Verify your email account"
- "Update your password"

Goal: Capture usernames, passwords, or OTPs.

6. Spear Phishing (Targeted Attacks)

Highly personalized emails using real information.

Example:

- Mentions your company, role, or recent activity
- Appears very convincing and relevant

Goal: Increase success by making the email look genuine.

7. Business Email Compromise (BEC)

Attackers pose as executives or finance staff.

Example:

- "Please transfer funds to this new account urgently"

Goal: Trick employees into sending money.

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

8. Lookalike Domains

Email addresses that are almost correct but slightly altered.

Example:

- finance@compaany.com instead of company.com

Defending Against Email and Browser Attacks

There are many actions that you can take to defend against email and browser attacks. Some of the most important ones are outlined here.

It is difficult to stop spam, but there are ways to reduce its effects:

- Most Internet service providers (ISPs) filter spam before it reaches the user's inbox.
- Many antivirus and email software programs automatically detect and remove dangerous spam from an email inbox.
- Organizations should educate employees about the dangers of unsolicited emails and make them aware of the dangers of opening attachments.
- Never assume that email attachments are safe, even when they come from a trusted contact. Always scan attachments before opening them.
- Become a member of the Anti-Phishing Working Group (APWG). It is an international association of companies focused on eliminating identity theft and fraud resulting from phishing and email spoofing.
- All software should be kept up-to-date, with the latest security patches applied to protect against any known security vulnerabilities
- Always verify the sender of an email before trusting its content
- Check the full email address carefully for any suspicious details
- Be cautious of unexpected or unusual requests, especially those involving money or sensitive information
- Avoid clicking on links without inspecting them first
- Hover over links to check their true destination before clicking
- When possible, navigate directly to official websites instead of using email links
- Handle email attachments with care
- Do not open attachments from unknown sources or those that seem out of context
- Be aware that attachments may contain malware designed to harm your system
- Never share sensitive information such as passwords, PINs, or one-time codes via email
- Remember that legitimate organizations do not request such information through email

FUTUREINTERNS-CYBERSECURITY INTERNSHIP

TASK-2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

- Ensure websites are secure by checking for HTTPS and the padlock symbol
- Avoid suspicious websites, pop-ups, and downloads from untrusted sources
- Keep browsers, operating systems, and applications up to date
- Install updates regularly to fix security vulnerabilities
- Use strong and unique passwords for different accounts
- Enable multi-factor authentication (MFA) for an added layer of security
- Organizations should implement spam filters, antivirus software, and access controls
- Regular employee training is essential to recognize and respond to threats
- Always pause before taking action on emails or online requests
- Think critically and verify information through trusted channels
- Be aware that many attacks rely on creating urgency to trick users

References

Jansson, K., & von Solms, R. (2013). Phishing for phishing awareness. *Behaviour & information technology*, *32*(6), 584-593.

Reinheimer, B., Aldag, L., Mayer, P., Mossano, M., Duezguen, R., Lofthouse, B., ... & Volkamer, M. (2020). An investigation of phishing awareness and education over time: When and how to best remind users. In *Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020)* (pp. 259-284)

Arshad, A., Rehman, A. U., Javaid, S., Ali, T. M., Sheikh, J. A., & Azeem, M. (2021). A systematic literature review on phishing and anti-phishing techniques. *arXiv preprint arXiv:2104.01255*.

Qabajeh, I., Thabtah, F., & Chiclana, F. (2018). A recent review of conventional vs. automated cybersecurity anti-phishing techniques. *Computer Science Review*, *29*, 44-55.